Cane vs. Synthetic: Evaluating the Material Science and Performance Realities of Clarinet Reeds

Michelle Cai

July 2026

Abstract

This study investigates the ongoing debate between traditional natural cane (Arundo donax) and modern synthetic clarinet reeds by bridging peer-reviewed material science data with autoethnographic playing experience. While synthetic alternatives like Légère reeds offer enhanced durability, predictability, and consistency over natural cane, significant physical and acoustical discrepancies remain. Through an analysis of existing literature on Young's modulus, holographic vibration mode frequencies, and compression hardness, this paper examines how synthetic materials differ in anisotropic behavior—particularly showing significantly higher transversal stiffness. These structural differences translate directly into performance challenges, including increased air resistance, delayed articulation attack (hysteresis), and a brighter, less flexible timbre in expressive passages. Ultimately, while synthetic reeds provide practical reliability, they cannot fully replicate the nuanced dynamic range, dark tone quality, and effortless responsiveness characteristic of premium cane reeds.

1 Introduction

I am a senior in high school, but I started playing the clarinet in fourth grade. I first learned to use the size 2.5 beginner Rico clarinet reeds and to set my embouchure, produce proper air support, and articulate. By middle school, I switched to size 3 Vandoren V12 reeds, blindly following the recommendation of my teacher. In the past year, I have noticed an increasing amount of clarinetists and wind players using plastic or synthetic reeds and this introduction to an unconventional material of clarinet reed sparked my curiosity. I began to wonder and ask myself questions: How does the unconventional material affect the intonation and tone of the clarinet? Given the growing number of clarinetists adopting the useage of synthetic reeds, what are the actual differences between synthetic and cane reeds in terms of their physical properties and playing experience?

The clarinet reed is an essential component for sound production in all clarinets. Typically made out of a cane called Arundo donax, reeds vibrate from air pressure to create the beautiful, rich sound of a clarinet. Leading clarinet reed brands include Vandoren, D'Addario, and Légère, all containing different cuts and shapes of the reeds and different sizes (reed thickness).

However, traditional cane reeds all have different properties due to the variable nature of the cane, and they also are subject to variations depending on the temperature, humidity, and weather. Additionally, they need to be broken in when first using it, played in rotation with other reeds as to avoid overuse, and replaced frequently to account for the degradation of the reed's fibers.

Due to these many drawbacks towards cane reeds, clarinetists are now searching for a more reliable and sustainable alternative. At first glance, synthetic reeds made of polymers and plastics offer clarinetists

the reliability, durability, and predictability that is desired. However, it is important to acknowledge the fundamental differences in sound production between cane and plastic reeds, and if these differences may hinder synthetic reeds from fully taking the place of cane reeds.

1.1 Intro to Reeds

Arundo donax thrives in warm, Mediterranean climates—particularly in southern France, Spain, and Italy. This plant is uniquely suited for reed making because of its internal structure: it has long, straight, parallel fibers that allow vibrations to travel evenly, and a naturally porous yet firm composition that balances flexibility with strength. These properties are essential for producing a stable, responsive tone when the reed vibrates against the clarinet's mouthpiece.

After harvesting, the cane undergoes a lengthy curing process, often lasting one to three years. This aging period is critical—it reduces moisture content, stabilizes the fibers, and prevents warping or cracking later on. Once cured, the cane is cut into tubes, split into sections, and then shaped into flat blanks. These blanks are carefully planed and profiled to precise thicknesses, as even tiny variations can significantly affect how a reed performs.

Before industrialization, reed making was an artisanal skill passed down through generations. In the 18th and early 19th centuries, individual musicians or highly trained craftsmen would hand-shape each reed using knives and simple tools. This meant that no two reeds were exactly alike, and players often had to adjust or refine reeds themselves to suit their embouchure and playing style.

That tradition still echoes in modern playing. Even with today's factory-made reeds, many musicians quickly learn that two reeds from the same box can respond very differently. Some feel too stiff, others too soft, and a few seem to "play perfectly" right away. Musicians like myself have to closely manage a careful rotation schedule, breaking in process, and even light sanding to obtain the ideal reed for each player.

Despite many technological advances since the 19th century, reed making still retains an element of unpredictability due to the natural variability of cane. As a result, many professional clarinetists continue to adjust their reeds by hand, fine-tuning aspects like tip thickness and vamp shape to achieve their desired sound and response. Others have begun to experiment with synthetic reeds, diverging from the longstanding tradition of cane reeds in hope for more reliability and longer durability.

2 Physical Parameters

An important measure of a reed's mechanical property is its Young's modulus, a measure of the stiffness of a solid material by telling us how much it stretches or compresses under a pull or push. The higher the value, the more rigid the material is. The Young's modulus can be split into a longitudinal modulus that measures along the grain of the reed and a transversal modulus, which measures across the grain. Traditional cane used to make reeds is anisotropic, meaning its longitudinal modulus is substantially greater than its transversal modulus [2].

2.1 Elasticity and Stiffness

Through experiments conducted by E. Ukshini et al. measuring the Young's modulus of different reeds, they discovered that the Légère reed came close to the properties of the cane material in the longitudinal direction, but was still almost 20 times stiffer than the wet cane in the transversal direction [4].

	Young's Modulus Transversal (GPa)	Young's Modulus Longitudinal (GPa)
Légère Reed	1.9(±0.1)	7.5(±0.3)
Dry Reed	0.5(±0.03)	5.0(±0.7)
Wet Reed	0.1(±0.05)	3.1(±0.4)

Table 1: **Summary of the moduli of elasticity** Above is data gathered by E. Ukshini et al. [4] that shows the measured moduli of a Légère reed compared to a dry and wet cane reed.

2.2 Range

K. Stetson et al. used holography to study physical differences between synthetic and cane reeds. By recording different vibration modes on a series of different kinds of reeds, their data showed that every synthetic reed tested had a lower fundamental vibration frequency than natural cane reeds (both wet and dry) as seen in Figure 1. These findings generate skepticism on the capability of synthetic reeds to replicate the exact mechanical properties of cane reeds.

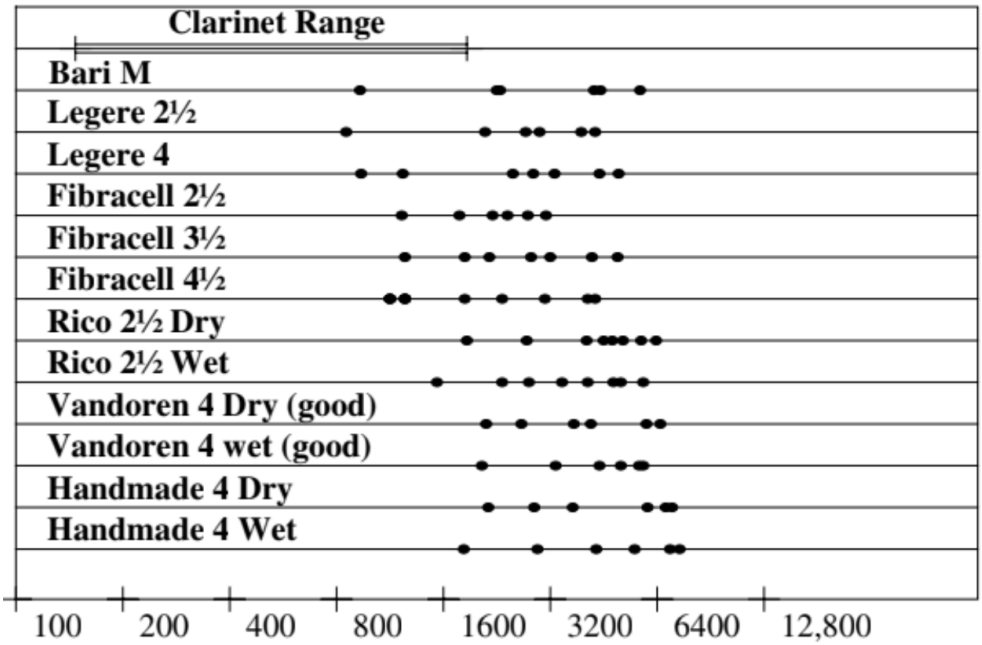


Figure 1: Tabulation of reed frequencies measured by K. Stetson et al. [3]

2.3 Hardness

Research conducted by M. Gangl et al. shows that synthetic reeds provide a greater resistance to displacement compared to cane reeds, making them harder to play on. In this experiment, M. Gangl et al. compressed the reeds at three different measuring positions, and recorded their data as shown in Figure 2.

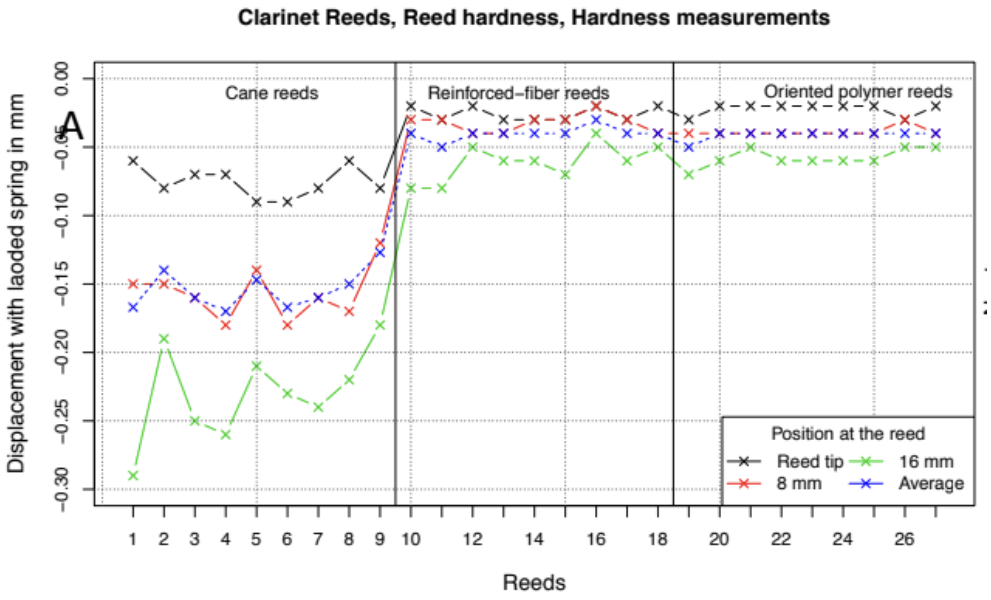


Figure 2: Hardness measurement of 27 different clarinet reeds via compression [1].

3 Acoustical Behavior and Playing Experience

To test the playing experience, I played different excerpts with each reed to compare their musicality. I started with holding long notes, increasing and decreasing the dynamic to get a sense of the dynamic ranges of each reed. In all three registers, bottom, middle, and top, I played pianissimo to fortissimo. Then, I played through a few major scales and a chromatic scale with different articulations, both legato and staccato, to experience the variations in playing experience in articulation. Finally, I played excerpts from three solo clarinet pieces: the third movement of Stravinsky's Three Pieces, Béla Kovács' Hommage a J.S. Bach, and the first movement of Brahms' first clarinet sonata. These three pieces are contrasting in style and technique. Stravinsky's Three Pieces requires rapid articulation with notes that are high in register, but less variation in dynamics. The Hommage à J.S. Bach, a contemporary etude, takes a slower tempo with more stylistic phrasing and dynamic contrast. Finally, the Brahms requires jumps between large intervals and a mixed variety of articulations. Playing on these pieces offers me a general understanding of the fundamental properties of and differences between the synthetic and cane reed.

3.1 Resistance

From my tests, I noticed that synthetic reeds generally have a greater resistance than a traditional cane reed. When playing and holding long notes, the Légère reed requires a noticeably greater amount of air support to produce the same dynamic of sound. As the register increases, the disparities between the resistances of the reeds become further accentuated, since the plastic reed requires a lot more air support for a note above a high C to come out. This disparity is also highlighted when playing in soft dynamics, as the sound produced by the synthetic reed sounds breathy in lower dynamics, incapable of mimicking the rich pianissimo of a cane reed.

Additionally, the fast articulation required in Stravinsky highlights fundamental differences in the resistances of the reeds. There is a slight hysteresis, or a delay, to every attack with the Légère reed, especially when there is a high register change. After the sound is produced, the sound lacks the clarity of a cane reed, especially in the middle register.

3.2 Timbre

Synthetic reeds overall have a brighter sound quality than cane reeds. This is more noticeable in slow, expressive pieces like the beginning of the Bach, as synthetic reeds struggle to imitate the rich tone of cane reeds. However, this difference in timbre is less noticeable and noteworthy when playing the Stravinsky because the piece is written so the clarinet naturally sounds bright and bouncy.

4 Conclusion

Ultimately, the documented playing experiences stem from the physical differences in the material composition. Due to the inability of the synthetic material to replicate the anisotropic behavior of the cane, synthetic reeds offer stronger resistance to vibration. While a cane reed is flexible transversally, a synthetic reed is much stiffer in the transversal direction, as explained previously, granting an explanation for the synthetic reeds to often require more air support to play the same note and dynamic. Similarly, the synthetic reeds generally have a slower response and attack than cane reeds due to the discrepancies between the transversal elasticity of the reeds.

Through this research process combining literary analysis of physical properties of cane and synthetic reeds with my personal playing experience as an advanced young musician, it is clear that synthetic reeds and cane reeds continue to have fundamentally different properties and behaviors. While synthetic reeds provide a consistency in sound that cane reeds are not able to achieve, synthetic reeds cannot imitate the foundational clear articulation and rich sound quality that cane reeds have for decades.

For developing young clarinetists who are interested in using synthetic reeds for their reliability, I believe that synthetic reeds like Légère's are best for pieces with a smaller dynamic range and slower tempo to minimize discrepancies with cane reeds.

References

[1] Manuel Gangl, Alex Hofmann, and Alexander Mayer. Comparison of characterization methods for b-flat clarinet reeds. In Mirko Čudina, editor, *Proceedings of the 7th AAAA Congress on Sound and Vibration*, pages 179–186, Ljubljana, Slovenia, 2016. Slovenian Acoustical Society.

[2] Guy Légère and Mark Kortschot. Oriented polymer reeds for woodwind instruments, March 8, 2011.

[3] Karl A. Stetson. Study of clarinet reeds using digital holography. *Optical Engineering*, 53(11):112305, 2014.

[4] Enis Ukshini and Joris Jan Jozef Dirckx. Longitudinal and Transversal Elasticity of Natural and Artificial Materials for Musical Instrument Reeds. *Materials*, 13(20):4566, 2020.